

SWMPP SECTION H

Annual Monitoring Plan for Fiscal Year 2021

July 1, 2020 – June 30, 2021

Daniel K. Inouye International Airport (HNL)
Small Municipal Separate Storm Sewer System
HI S000005



Prepared For:

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Version 7

RECORD OF REVISION

Revision No.	Revision Date	Description	Sections Affected
1	May 2007	Version 1.0 – Initial Release	All
2	June 2009	Version 2.0 – Clarification on monitoring locations, parameters, and frequency	2.0, App A
3	June 2011	Version 3.0 – NPDES permit extension	1.2, 2.1, 2.2, 2.3
4	April 2014	Version 4.0 - Revised to comply with requirements of the Annual Monitoring Plan per the revised HIS000005	All
5	May 2015	Version 4.1 – Updated for 2015	3.8, 4.4
6	April 2016	Version 4.2 – Updated for 2016	2.0, 4.4
7	April 2017	Version 4.3 – Updated for 2017	4.4
8	April 2018	Version 5 – Updated for Fiscal Year 2019 to include automatic sampler procedures and updates to effluent limitations	All
9	April 2019	Version 6 – Updated for Fiscal Year 2020	All
10	February 2020	Version 6.1 – Amendment to Version 6. Updated to remove HNL 002 monitoring location.	All
11	April 2020	Version 7 – Updated for Fiscal Year 2020	All

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

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LIST OF ACRONYMS

AOA	Airport Operations Area
BMPs	Best Management Practices
CFR	Code of Federal Regulations
COC	Chain of Custody
DMR	Discharge Monitoring Report
DOH	State of Hawaii, Department of Health
DOH-CWB	State of Hawaii, Department of Health, Clean Water Branch
DOTA	State of Hawaii, Department of Transportation, Airports Division
HAR	Hawaii Administrative Rules
HNL	Daniel K. Inouye International Airport
EID	Environmental Identification Number
EPA	U.S. Environmental Protection Agency
GPM	gallons per minute
I&M	Implementation and Monitoring
IDDE	Illicit Discharge Detection and Elimination
MDL	Method Detection Limit
MS4	Municipal Separate Storm Sewer System
NPDES	National Pollutant Discharge Elimination System
OWS	Oil Water Separator
PPE	Personal Protective Equipment
QA/QC	Quality Assurance/Quality Control
RL	Reporting Limit
SPCC	Spill Prevention, Control, and Countermeasure
SWMPP	Stormwater Management Program Plan
SWPCP	Stormwater Pollution Control Plan
TMDL	Total Maximum Daily Load
WLA	Waste Load Allocation

1.0 INTRODUCTION

1.1 Background

The State of Hawaii, Department of Transportation, Airports Division (DOTA) owns and operates a Small Municipal Separate Storm Sewer System (MS4) at Daniel K. Inouye International Airport (HNL). The State of Hawaii, Department of Health, Clean Water Branch (DOH-CWB) issued a National Pollutant Discharge Elimination System (NPDES) Individual Permit, No. HI S000005 for HNL (hereinafter Permit), which authorizes DOTA to discharge stormwater runoff and certain non-stormwater discharges from its Small MS4 as well as stormwater runoff from DOTA municipal industrial facilities (i.e., Baseyards) into State Waters. The Permit became effective on April 14, 2014 and expired on March 13, 2019. DOH-CWB administratively extended the Permit on March 1, 2019, and the administrative extension is set to expire on the effective date of the subsequent Permit.

This Annual Monitoring Plan is a part of the HNL Small MS4 Stormwater Management Program Plan (SWMPP) as Section H and will serve to assess compliance with the terms of Permit Part F. The focus of this program is to monitor stormwater runoff from the HNL Small MS4, including stormwater discharges from industrial facilities which discharge to state waters.

Pursuant to Permit Part F.1.a, this Annual Monitoring Plan covers monitoring activities planned for the fiscal year (FY) 2021, from July 1, 2020 through June 30, 2021. Pursuant to Permit Part A.6, the Annual Monitoring Plan will be available on DOTA's website for public review and comment for 30 calendar days, and after comments are addressed, submitted to DOH-CWB by June 1st each year.

Note that HNL 002 was removed as a monitoring location, since it is not a requirement of the permit. Refer to Annual Monitoring Plan, Version 6.1 for more information on this amendment. HNL 003 remains as the industrial activities monitoring location for HNL.

1.2 Objectives

The purpose of this Annual Monitoring Plan is to document proposed objectives and provide a description of DOTA's monitoring activities to be performed within the 2021 fiscal year. The goal of the monitoring program is to allow DOTA and DOH-CWB to assess the effectiveness of existing stormwater management procedures.

Pursuant to Permit Part F.1.a, F.1.b.(1), and F.1.b.(2), the Annual Monitoring Program includes the following objectives and implementation measures to meet these objectives. The Annual Monitoring Report will include an evaluation of these objectives (see Section 4.2.2).

1.2.1 Objective #1: Compliance

Monitoring will be conducted to assess compliance with the Permit. The Annual Report provides a comprehensive assessment of compliance each year. Additionally, compliance may also be evaluated in accordance with Total Maximum Daily Load (TMDL) Implementation and

Monitoring (I&M) Plans and to demonstrate consistency with Waste Load Allocations (WLAs). However, TMDLs and WLAs have not been assigned to the receiving waters at HNL at this time.

1.2.2 Objective #2: SWMPP Effectiveness

Monitoring will be conducted to measure the effectiveness of the HNL SWMPP with regards to the programs that may be implemented in the monitoring area. Specific programs are further described in Section 1.3. Pursuant to Permit Part G.1.d, DOTA submitted a Program Effectiveness Strategy (June 2015) to DOH-CWB. The Program Effectiveness Strategy includes indicators of program effectiveness and notes that SWMPP Sections A through G include measurable standards and milestones for each component of the program. The Annual Report details the progress made on these standards and milestones, and is used to assess effectiveness of the SWMPP.

1.2.3 Objective #3: Overall Health

Monitoring will be conducted to assess the overall health of the runoff based on chemical, physical, and biological impacts to receiving waters. As the data is compiled from sampling events, long-term trends will be evaluated so that the HNL SWMPP can be targeted to specific areas of concern.

1.2.4 Objective #4: Characterization

Monitoring will be conducted to characterize stormwater discharges leaving the HNL Small MS4, including DOTA industrial facilities. Field observations will supplement discharge characterization from monitoring.

1.2.5 Objective #5: Pollutant Source Identification

Monitoring will be conducted to identify specific pollutant sources. Once identified, the pollutant sources can be more readily targeted with operational Best Management Practices (BMPs) or Permanent BMPs (low impact development, source control, or treatment devices).

1.2.6 Objective #6: Illicit Discharge Detection and Elimination (IDDE)

Monitoring will be conducted to assist the IDDE Program, which is designed to prevent illegal discharges and/or connections to the HNL Small MS4. Pursuant to Permit Part D.1.c.(2), DOTA submitted an Outfall Inspection and Field Screening Plan (October 2014) to DOH-CWB. This screening plan describes DOTA's process to conduct outfall inspections and investigate illicit discharges. Once an illicit discharge has been identified, DOTA will make an immediate effort to stop the discharge and prevent it from recurring. In fiscal year 2021, DOTA may incorporate water quality sampling for source identification if the typical source tracing methods are unsuccessful. Additionally, DOTA is developing an SOP for IDDE investigations that includes sampling for any observed and unknown illicit discharges.

1.2.7 Objective #7: Assess Receiving Water

Monitoring will be conducted to assess how runoff from HNL may be contributing to water quality issues in the receiving waters (Kaloalua Canal, Keehi Lagoon and the Pacific Ocean). DOTA will assess impacts to receiving waters and specific water quality issues resulting from DOTA MS4 discharges.

1.3 Management Measures

Per Permit Part F.1.b.(3), DOTA has analyzed its management measures for effectiveness at reduction of pollutants and flow. Monitoring location HNL 003 is representative of industrial activities at the Maintenance Baseyard. Effectiveness of the management measures at the Maintenance Baseyard, including personnel training program, implementation of BMPs, spill response practices, the debris control program, adherence to the Maintenance Baseyard's Stormwater Pollution Control Plan (SWPCP), facility inspections, and the permanent BMP program are described below. Also described are mitigation measures in response to sampling exceedances.

Education and training programs are an important management measure for stormwater quality. DOTA conducts annual training at the Maintenance Baseyard for several topics, including but not limited to: spill response (including a spill drill); implementing their SWPCP and Spill Prevention, Control, and Countermeasure (SPCC) Plan; general stormwater protection; and chemical application BMPs training. These trainings continue to be an effective measure. The annual training includes instruction on BMPs specific to the Maintenance Baseyard, which are summarized in the Maintenance Baseyard Factsheet (a handout DOTA made specifically for the facility). Maintenance personnel are receptive to the training and knowledgeable on BMPs. DOTA has found that once maintenance personnel are aware of their stormwater impacts and the BMPs that they can implement, they begin to incorporate stormwater pollution prevention into their decision-making process.

In addition to the formal classroom setting training sessions and handouts, DOTA provides additional training during semi-annual facility inspections. The facility inspections document deficiencies, which the Baseyard is required to correct within 30 days. Along with correcting deficiencies at the site, these inspections identify areas that the Maintenance Baseyard could improve upon so that DOTA can enhance training on those topics.

DOTA has a trash reduction plan that begins with a three-year baseline study. A baseline boom on Kaloaloa Canal downstream from the HNL 003 sampling location was installed in January 2019. The effectiveness of these baseline booms and the trash reduction plan in general will be evaluated in the upcoming Annual Report.

Managing storm drain inlet cleaning and street sweeping is vital to minimize pollution to DOTA's stormwater. Storm drain inlet inspection and maintenance, including canal and shoreline cleaning, is conducted as necessary by a service contractor. A new Inspection, Maintenance, and Pollution Prevention of MS4 Contract (BS1927-23) which has been awarded to Eckard Brandes (with Hawaiian Industrial Services, LLC as a subcontractor), began in September 2019, and includes drain cleaning, canal cleaning, OWS cleaning, and cleaning of certain PBMPs. Street sweeping is conducted at least twice a month by the Maintenance Baseyard. Storm drain inlet cleaning and street sweeping helps remove debris and sediment before they can impact receiving waters.

In response to monitoring exceedances, DOTA increased the street sweeping frequency at the Maintenance Baseyard in fiscal year 2020. The Maintenance Baseyard will be swept more frequently than twice a month when feasible.

Permanent BMPs reduce pollutants and provide water quality improvements. There are three drains at the Maintenance Baseyard near the fueling area that have Safe Drains® installed (Environmental Identification Numbers [EIDs] 5502, 5503 and 5504). These inlet systems can be closed quickly in the event of a spill and are effective at minimizing oil and grease in stormwater.

DOTA installed five drain inlet filter units with multi-layer filter cartridges at the Maintenance Baseyard in November 2017. The five drains with their corresponding inlet insert PBMPs are as follows: drain 5499 with insert 4910, drain 5500 with insert 4911, drain 5502 with insert 4912, drain 5503 with insert 4913, and 5504 with insert 4914. They were installed as part of HNL's Retrofit Action Plan. The filter media in these units is designed to treat dissolved and particulate metals. These drains also have a fitted boom to further aid in absorbing and filtering metals.

Unfortunately, since the installation of the five drain inlet filters, sampling results have had exceedances of metals. In response, DOTA noticed that the covered parking structure near drain inlet EID 5499 was severely rusting, resulting in metal rust flakes accumulating on the ground. On March 27, 2019, DOTA implemented a temporary source control BMP by sanding and then painting the metal surfaces with corrosion inhibiting paint, removing the corroded fencing which will be replaced with new fencing, and sweeping the area to remove the rust flakes (see Figures 1 - 3). This is a temporary measure until a capital improvement project can be planned to refurbish the structure.



FIGURE 1. RUSTY COVERED PARKING STRUCTURE (MARCH 13, 2019).



FIGURE 2. COVERED PARKING STRUCTURE MITIGATED (MARCH 27, 2019). Metal surfaces painted with corrosion inhibiting paint. Corroded fencing removed. Rust flakes on the ground swept up.



FIGURE 3. COVERED PARKING STRUCTURE MITIGATED (APRIL 15, 2019). Fencing replaced.



FIGURE 4. GR8 GUARD DRAIN PROTECTOR MAT AT DRAIN INLET 5499 (MAY 6, 2019).

On May 6, 2019, DOTA installed another temporary BMP, a drain protector mat (specifically a GR8 Guard) at drain inlet EID 5499, the inlet closest to the severely rusted parking structure (refer to Figure 4). While the rust problem on the covered parking structure was temporarily mitigated, this drain protector mat will provide another level of protection to capture any new rusted metal flakes. It may also assist in combating the nutrients and turbidity exceedances by protecting against possible loose soil and fertilizer from the greenhouse or aggregate storage area.

Manhole cover EID 5501 was observed to be rusty. The manhole cover was treated and painted in July 2019 (refer to Figure 5). Manhole EID 5505 was also observed to be rusty, but EID 5501 was more pertinent.



FIGURE 5. MANHOLE 5501 PAINTED (JULY 2019).

DOTA has begun a construction project for the Heavy Equipment Garage at the HNL Maintenance Baseyard (AO1142-15). Construction began in November 2019, delayed from the original anticipated date of September 2019. The construction delay was due to issue with the manufacturer for the structural components. The anticipated Project Construction Completion date is estimated

to be December 2020. The new Heavy Equipment Garage will provide shelter for heavy equipment, thus lessening the potential impact to stormwater.

Additionally, maintenance of the five drain inlet inserts, which involves replacing the filter media and the booms, was conducted under the Inspection, Maintenance, and Pollution Prevention of MS4Contract (BS1927-23) on December 11, 2019. Refer to Figures 6 – 13 for example photos of one the inlet inserts media replacement. Approximately 200 pounds (1.8 cubic feet) of treatment media was replaced in each of the five inlet inserts. No debris was present in each of the inlets. Under the Inspection, Maintenance, and Pollution Prevention of MS4 Contract (BS1927-23), DOTA has the ability to authorize the contractor to replace the filter media more frequently, so that pollutants that could potentially leach from the filter media do not affect the stormwater and to ensure proper maintenance of these drain inlet inserts.



**FIGURE 6. SAFE DRAIN OF INLET 5503
BEFORE CLEANING**



**FIGURE 7. DRAIN INLET INSERT 4913 (IN
INLET 5503) BEFORE CLEANING**



FIGURE 8. INTERIOR OF INLET 5503



**FIGURE 9. OLD MEDIA OF INSERT 4913 TO BE
REMOVED**



FIGURE 10. OLD BLUE FILTER OF INSERT 4913 TO BE REMOVED



FIGURE 11. CLEAN MEDIA INSERTED INTO 4913



FIGURE 12. NEW BLUE FILTER INSERTED INTO 4913



FIGURE 13. NEW ABSORBENT BOOM ATTACHED IN 4913

DOTA hopes that the combination of these management measures will help reduce stormwater sampling exceedances. DOTA will continue to sample and monitor stormwater and will use an adaptive management approach to evaluate other potential BMPs/PBMPs if necessary.

Further evaluation of HNL SWMPP effectiveness may be found in the HNL Annual Monitoring Report and the HNL Annual Report.

1.4 Updates from Previous Annual Monitoring Plan

Changes between this year's FY 2021 Annual Monitoring Plan and the FY 2020 Annual Monitoring Plan Amendment (Version 6.1) include:

- Updates to Section 1.3 for additional management measures implemented over the year in response to monitoring exceedances;
- Updated figures in Appendix A;
- Updated logo, acronyms and other minor edits throughout.

2.0 MONITORING PROGRAM

The stormwater monitoring program at HNL includes one sample location. The industrial activities monitoring location (HNL 003) provides an analysis of discharge from the Maintenance Baseyard. Appendix A includes figures of the monitoring location, including Figure 1 – Site Location Map, Figure 2 – Site Map, and Figure 3 – Stormwater Flow Chart.

2.1 Industrial Activities Monitoring Location (HNL 003)

Stormwater runoff from the DOTA Maintenance Baseyard located at 2919 Aolele Street, Honolulu, Hawaii is monitored at HNL 003 to capture a representative sample of impacts from industrial activities. An automatic sampler was installed at this sample location in October 2017. In May 2018, the monitoring location (including configuration of the automatic sampler) was adjusted to capture the sample from the pipe in between manhole 5501 and oil water separator (OWS) 9363, instead of outfall 4576. This relocation was based off of observed tidal influence at the outfall and the OWS, but not at the manhole. Hence DOTA adjusted the monitoring location to provide a more accurate representation of stormwater runoff from the Maintenance Baseyard. For more information on the sampling point relocation, refer to the April 2018 Annual Monitoring Plan. Other software and hardware upgrades have also been made to the autosampler over the course of the year.

- **HNL 003** (in between Manhole 5501 and OWS 9363, Maintenance Baseyard) captures runoff from multiple drain inlets located in the Maintenance Baseyard. Stormwater discharge flows to the Kaloaloa Canal, which is considered a Class 2 Inland Water with salt water intrusion from Keehi Lagoon, a Class A Marine Water. Effluent limitations for brackish/saline estuary inland waters (HAR Chapter 11-54-5.2(d)(1)) and salt water (Permit, Part F.2, note 11 & 7) will be applied. An automatic sampler with a multiparameter probe is located at this monitoring location.

TABLE 1: INDUSTRIAL ACTIVITIES MONITORING LOCATION – HNL 003

STORM DRAIN NUMBER	GPS COORDINATES	DESCRIPTION OF RUNOFF	POTENTIAL POLLUTANTS
In between Manhole 5501 and OWS 9363	21° 19' 53.11" N 157° 54' 21.71" W	Maintenance Baseyard vehicle maintenance, washing, and fueling, storage areas for maintenance materials and waste	Diesel / gasoline, lubricants / oils, VOCs, heavy metals, pesticides, surfactants, debris, and hazardous waste

2.2 Sampling Frequency

Stormwater sampling shall be conducted annually for the parameters identified in Table 2, including any additional parameters that HNL suspects may be present in the stormwater. In accordance with HAR Chapter 11-55, Appendix B, if there is an exceedance, sampling will be conducted for the next representative storm event until all parameters are met, unless informed otherwise by the DOH Director.

2.3 Monitoring Parameters

The parameters to monitor are in Table 2. Note that the automatic sampler at HNL 003 has a field sonde. Monitoring shall be conducted in accordance with HAR Chapter 11-55 Appendix B. Effluent limitations for brackish/saline estuary inland waters (HAR Chapter 11-54-5.2(d)(1)) and salt water (Permit Part F.2, note 11 & 7) are applied. Monitoring test methods shall be conducted in accordance with Code of Federal Regulations (CFR), Title 40, Part 136. Method Detection Limits (MDLs) shall be below the effluent limits. Note that test methods may be subject to change, but any method used will be a 40 CFR 136 approved method (1664A only will be used for oil and grease per the Permit). MDLs and reporting limits (RLs) will be updated per standard laboratory practice and are subject to change based on lab, method and/or instrumentation.

TABLE 2: MONITORING PARAMETERS

PARAMETER (UNIT)	SAMPLE TYPE ²	EFFLUENT LIMITATION ¹	TEST METHOD	RL	MDL
Flow (gpm)	Calculate/ Estimate	Report ⁴	Calculate/ Estimate (Field)	--	--
pH (unit)	Grab ⁹	5.5-8.0	AQ 702 Sonde field readings	--	--
Dissolved Oxygen (mg/l)	Grab	Report ⁴	AQ 702 Sonde field readings	--	--
Oxygen Saturation (%)	Grab	≥ 75%	AQ 702 Sonde field readings	--	--
Temperature (°C)	Grab	±1 °C from ambient*	AQ 702 Sonde field readings	--	--
Salinity (ppt)	Grab	10% from ambient*	AQ 702 Sonde field readings	--	--
Oil and Grease (mg/l)	Grab	15 mg/l	EPA 1664A	5 mg/l	1.4 mg/l
Turbidity (NTU)	Grab	3.00 NTU	EPA 180.1	0.01 NTU	--
Biochemical Oxygen Demand (5-Day) (mg/l)	Composite ³	Report ⁴	SM 5210B	1 mg/l	--
Chemical Oxygen Demand (mg/l)	Composite ³	Report ⁴	EPA 410.4	20 mg/l	10 mg/l
Total Suspended Solids (mg/l)	Composite ³	Report ⁴	SM 2540D	0.1 mg/l	--
Total Phosphorus (mg/l)	Composite ³	0.05000 mg/l	SM 4500-P-B-5-E	0.02 mg/l	0.01 mg/l
Total Nitrogen (mg/l) ⁵	Composite ³	0.35000 mg/l	Calculation (EPA 351.2 for Total Kjeldahl Nitrogen (TKN))	--	0.05 mg/l

PARAMETER (UNIT)	SAMPLE TYPE ²	EFFLUENT LIMITATION ¹	TEST METHOD	RL	MDL
Nitrate + Nitrite Nitrogen (mg/l)	Composite ³	0.02500 mg/l	SM 4500-NO3-F	0.05 mg/l	0.02 mg/l
Ammonia Nitrogen (mg/l)	Composite ^{3~}	0.01000 mg/L	SM 4500-NH3-G	0.1 mg/l	0.01 mg/l
Aluminum (µg/l) ¹²	Composite ³	Report ⁴	EPA 200.8	10 µg/l	5 µg/l
Cadmium (µg/l) ¹²	Composite ³	43 µg/l	EPA 200.8	1 µg/l	0.25 µg/l
Chromium (VI) (µg/l) ¹²	Composite ³	1,100 µg/l	SM 3500-Cr-B	20 µg/l	--
Copper (µg/l) ¹²	Composite ³	2.9 µg/l	EPA 200.8	2 µg/l	0.5 µg/l
Lead (µg/l) ¹²	Composite ³	140 µg/l	EPA 200.8	1 µg/l	0.5 µg/l
Nickel (µg/l) ¹²	Composite ³	75 µg/l	EPA 200.8	2 µg/l	0.5 µg/l
Silver (µg/l) ¹²	Composite ³	2.3 µg/l	EPA 200.8	1 µg/l	0.5 µg/l
Zinc (µg/l) ¹²	Composite ³	95 µg/l	EPA 200.8	20 µg/l	2.5 µg/l
Benzene (µg/l)	Composite ³⁺	1,700 µg/l	EPA 624.1	0.5 µg/l	0.065 µg/l
Additional Toxic Pollutants ¹³ (None currently)	Grab or Composite ¹⁵	Report ¹⁴	--	--	--

NOTES:

EPA = U.S. Environmental Protection Agency

gpm = gallons per minute

mg/l = milligrams per liter = 1000 micrograms per liter (µg/l)

MDL = Method Detection Limit

NTU = Nephelometric Turbidity Units

RL = Reporting Limit

* = Monitoring ambient conditions is currently out of DOTA's capabilities.

~ = Footnote #3 was not applied to this parameter in the Permit, but DOTA infers that this composite sample will be handled the same as the other composite samples.

+ = The laboratory will composite using appropriate procedures for volatile compositing.

¹ Pollutant concentration levels shall not exceed the stormwater discharge limits or be outside the ranges indicated in the table. Actual or measured levels which exceed those stormwater discharge limits or are outside those ranges shall be reported to the DOH-CWB required in HAR, Chapter 11-55, Appendix B, Section 10(c).

² The Permittee shall collect samples for analysis from a discharge resulting from a representative storm. A representative storm means a rainfall that accumulates more than 0.1 inch of rain and occurs at least 72 hours after the previous measurable (greater than 0.1 inch) rainfall event.

“Grab sample” means a sample collected during the first 15 minutes of the discharge.

“Composite sample” means a combination of at least two (2) sample aliquots, collected at periodic intervals. The composite shall be flow proportional; either the time interval between each aliquot or the volume of each aliquot must be proportional to the total flow of stormwater discharge flow since the collection of the previous aliquot. The Permittee may collect aliquots manually or automatically.

Samples for analysis shall be collected during the first 15 minutes of the discharge and at 15-minute intervals thereafter for the duration of the discharge, as applicable. If the discharge lasts for over an hour, sample collection may cease.

³ If the duration of the discharge event is less than 30 minutes, the sample collected during the first 15 minutes of the discharge shall be analyzed as a grab sample and reported toward the fulfillment of this composite sample specification. If the duration of the discharge event is greater than 30 minutes, the Permittee shall analyze two (2) or more sample aliquots as a composite sample.

⁴ Monitor and Report. The value shall not exceed the applicable limit as specified in Chapter 11-54 for the applicable classification of the receiving state waters. If no limitation is specified in Chapter 11-54, then the Permittee shall monitor and report the analytical result. The Department may include discharge limitations specified in Section 11-55-19 and discharge limitations based on Federal Register, Vol. 73, No. 189, Pages 56572–56578, dated September 29, 2008.

⁵ **Total Nitrogen is a measure of all nitrogen compounds in the sample** (nitrate, nitrite, ammonia, dissolved organic nitrogen, and organic matter present as particulates).

⁹ The Permittee shall measure pH within 15 minutes of obtaining the grab sample.

¹² The Permittee shall test for the total recoverable portion of all metals.

¹³ Toxic pollutants, as identified in Appendix D or 40 CFR Part 122 or in HAR, Chapter 11-54, Section 11-54-4, need only be analyzed if they are identified as potential pollutants requiring monitoring in the SWPCP. The Permittee shall test for the total recoverable portion of all metals. If monitoring results indicate that the discharge limitation was equaled or exceeded, the SWPCP shall be amended to include additional BMPs targeted to reduce the parameter which was in excess of the discharge limitation.

¹⁴ Effluent limitations are the acute water quality standards established in HAR, Chapter 11-54, Section 11-54-4. For pollutants which do not have established acute water quality standards, any detection concentration greater than 0.01 mg/l shall be reported.

¹⁵ Cyanide and the volatile fraction of the toxic organic compounds shall be sampled by grab sample. All other pollutants, as identified in Appendix D of the 40 CFR Part 122 or in HAR Chapter 11-54, Section 11-54-4 shall be sampled by composite sample.

3.0 WATER QUALITY MONITORING GUIDE

This section describes the procedures that will be followed when collecting stormwater samples and the Quality Assurance (QA)/Quality Control (QC) procedures.

3.1 Sample Collection Preparation

The automatic sampler at HNL 003 has reusable bottles that are dedicated to the sampling site to prevent cross-contamination. Sampling personnel shall coordinate with the laboratory to ensure that the automatic sampler bottles are decontaminated following the laboratory's internal QA/QC procedures for cleaning equipment.

Sampling personnel shall coordinate with the laboratory to ensure that adequate sample volume is collected for analysis of the parameters listed in Table 2.

The automatic samplers are configured to send text message alerts to designated personnel at the start of a representative storm and sample collection. The following equipment and materials are suggested to be readily on hand:

- Sample cooler with ice.
- Field notebook (a rain resistant notebook is suggested) and/or Stormwater Sampling Logs (Appendix C), sample labels, chain of custody (COC) forms, and a permanent marking pen.
- Disposable nitrile gloves. Note: these should be worn when handling samples and containers.
- Personal Protective Equipment (PPE) such as AOA badge, safety vest, safety glasses, steel toed boots, etc., as well as a car with an AOA sticker and driver with a ramp driving permit.
- A second set of automatic sampler bottles (if available, to replace the existing ones).

3.2 Automatic Sampler Collection

DOTA will attempt to obtain samples between Monday and Friday morning to account for applicable laboratory hold times.

The automatic sampler is configured to only sample during a representative storm event, i.e. rainfall that accumulates more than 0.1 inches of rain and occurs at least 72 hours after the previous 0.1 inch rainfall event. Rainfall will be measured and logged by the integrated rain gauge at the autosampler. An area-velocity (AV) sensor will record the flow rate. A field probe will record some of the grab sample parameters (see Table 2).

Within the first 15-minutes of the storm event, the automatic sampler will collect a sample for all parameters listed in Table 2 (grab and composite), excluding those parameters that will be measured in the field by the probe. Thereafter, the automatic sampler will continue collecting the composite samples at 15-minute intervals for a duration up to one hour, or earlier if the discharge stops. The parameters will be analyzed as a grab sample if the discharge ends after only one

sample has been collected. If two or more samples were collected, the laboratory will combine the composite samples appropriately.

All sample bottles should be labeled. An example is provided below.

Date: 1/1/15	Time: 0900	Collected By: J. Smith
Sample Name: 1_HNL003		Sampling Site: HNL 003
Tests Required: See COC		
Sample Type: ☐ Grab ☒ Composite ☐ Other_____		

FIGURE 14. EXAMPLE OF A SAMPLE BOTTLE LABEL

A COC form will be completed and signed by each individual handling the samples to ensure sample integrity. The samples will be transported to the laboratory in a cooler filled with ice. The automatic sampler bottles will be replaced immediately with a second, clean set, or after the existing set is cleaned.

3.3 Field Notes

The following observations will be recorded in the field notebook and/or on a Stormwater Sampling Log (Appendix C):

- Monitoring location.
- Name(s) of sampling personnel.
- Date and time of arrival onsite.
- Probe calibration date.
- Automatic sampler activation times.
- Storm event characteristics (duration, time, magnitude (total rainfall), and date of last rain event greater than 0.1 inches).
- Water quality observations of the discharge (by looking at the sample bottles), such as the following: color, odor, clarity, floating solids, settled solids, suspended solids, foam, oil sheen, or other obvious indicators of stormwater pollution.
- Field monitoring parameters (such as pH, temperature, salinity, dissolved oxygen, and flow rate).
- Samples collected (total number and individual sample information).

3.4 Quality Assurance/Quality Control (QA/QC)

The testing laboratory shall be qualified to perform the approved analytical/test methods for analyzing environmental samples per 40 CFR 136. The methods for analyzing environmental samples contain explicit quality control requirements that must be met. These requirements include

specific procedures and criteria for evaluating accuracy and precision, demonstrating the ability of the analyst to generate acceptable accuracy and precision, and demonstrating that extraneous interferences are under control. The laboratory will be required to document strict adherence to the general laboratory QA/QC requirements. The laboratory shall provide appropriate QA/QC documentation with the analytical results. The laboratory results should include the date(s) the analyses were performed, the individual(s) who performed the analyses, the analytical techniques or methods used, the MDL, the RL, the results of the analyses, and QA/QC documentation.

Field QA/QC includes cleaning of the automatic sampler containers. At a minimum, this involves using phosphate-free detergent and rinsing with DI or distilled water three times. The laboratory will clean the automatic sampler containers, following their internal laboratory QA/QC procedures for cleaning equipment. The analysis of rinsate blank samples may be conducted as appropriate to ensure that the containers are properly decontaminated.

The field probe at HNL 003 will be calibrated per the manufacturer's guidelines. A calibration log can be exported from the software and stored electronically.

3.5 Data Review and Validation

Upon receipt of the analytical results from the laboratory, personnel will perform additional data validation to determine whether analytical data is acceptable for use. The evaluation will include an assessment of laboratory QA/QC data and field notes. The field measured parameters will be validated by the calibration of the water quality probe. Rinsate blank sample analysis may be conducted to test decontamination of the automatic sampler containers.

4.0 ANALYSIS AND REPORTING

4.1 Data Analysis

The analytical results of the samples will be used to determine compliance with the Permit. Should effluent limitations be exceeded, an assessment will be conducted to determine possible sources for that exceedance. Once identified, additional measures may be taken to further contain the pollutants.

Monitoring location HNL 003 is representative of industrial activities at the airport Maintenance Baseyard. Therefore, the sample is representative of the effectiveness of the Baseyard personnel training program, the implementation of BMPs, spill response practices, the debris control program, and adherence to the Maintenance Baseyard SWPCP. Should effluent limitations be exceeded, these programs may be revised to prevent the discharge of pollutants.

4.2 Reporting Requirements

Once the laboratory provides the sample analysis results, DOTA must evaluate those findings and communicate them to interested parties.

4.2.1 DMRs

DOTA will complete DMRs to document the sample findings in comparison to the effluent limitations listed in Table 2. A DMR template for HNL 003 is included in Appendix B. Each exceedance of the effluent limitations will be specifically noted with a plan for correction (see Section 4.2.3). If a test result is not detectable, the result shall be reported as “less than #” where the # is the lowest detection limit of the test method (per HAR Chapter 11-55, Appendix B, Section 8.a.4.C). If sampling did not occur for the fiscal year, a DMR will be submitted that includes an explanation as to why sampling was not conducted (e.g. there was no discharge).

In addition to the DMR form, the DMR submittal shall include a cover letter, the laboratory report with QA/QC data, as well as the flow, the start and end time of the monitored storm event, and the duration since the last storm event of 0.1 inch or more (see Appendix C for a stormwater sampling log template).

Completed DMRs should be signed by the Director of Transportation and submitted no later than 60 (sixty) days following the sample collection via the e-Permitting portal or NetDMR, once available. The DOH-CWB website provides DMR instructions.

4.2.2 Annual Monitoring Report

The Annual Monitoring Report is due by August 31st each year to the DOH-CWB. It covers activities conducted during the past fiscal year and shall, at a minimum, include the following items per Permit Part G.2.b:

- a) Discussion on the activities/work implemented to meet each objective as outlined in Section 1.2, including any additional objectives identified as well as the results and conclusions;

- b) Written narrative of the past fiscal year's activities, including those coordinated with other agencies, objectives of activities, results and conclusions;
- c) Data gathered on levels of pollutants in non-stormwater discharges to the HNL Small MS4;
- d) Using rainfall data collected, relate rainfall events, measured pollutant loads, and discharge volumes from the watershed;
- e) Date when monitoring occurred and the results.
- f) DMRs.

4.2.3 Reporting Exceedances

For each exceedance of the effluent limitations, an oral report shall be made to DOH-CWB via telephone (808) 586-4309 during normal business hours (Monday through Friday from 7:45 am until 4:15 pm, excluding holidays), as soon as the results become available, detailing the parameters exceeded and the dates and times. For after business hours, the non-compliance may be reported to the Hawaii State Hospital Operator (808) 247-2191.

In addition to the oral report, a written report shall be included in the DMR. The written report shall include a description of the exceedance(s) and suspected origin, the dates and times, estimation of how long the exceedance is expected to continue, and measures planned or implemented to combat the exceedance. During the oral report, it shall be requested to submit the written report with the DMR (versus the five (5) day period specified in HAR 11-55 Appendix B), in order to provide sufficient time to obtain DOTA Director signatures.

4.3 Budget

The proposed budget for fiscal year 2021 is \$30,000 to collect, analyze, and report the findings of the stormwater sampling, as well as routine maintenance of the autosamplers and upgrading autosampler parts as needed.

5.0 WASTELOAD ALLOCATION

Currently, HNL does not have any TMDLs assigned to its receiving waters or WLAs identifying it as a point source at this time. As WLAs are adopted by DOH, Environmental Planning Office, that identify DOTA as a source, DOTA shall develop TMDL I&M Plans for a minimum of one (1) additional TMDL per year within one (1) year of the adoption date. Each I&M Plan shall include a schedule that: has a final deadline for the WLAs consistent with the TMDL document; adheres to HAR, Section 11-55-21 and 40 CFR 122.2 and 122.47; includes implementation of BMPs; includes monitoring to evaluate performance; and provides time/flexibility to make changes needed to meet the WLAs. If the schedule is longer than one year, it shall include interim dates and milestones that do not lapse over one year.

6.0 REFERENCES

State of Hawaii, Department of Health. November 2014. *Hawaii Administrative Rules, Chapter 11- 54.*

State of Hawaii, Department of Health. December 2013. *Hawaii Administrative Rules, Chapter 11- 55, Appendix B.*

State of Hawaii, Department of Health. February 2019. *Hawaii Administrative Rules, Chapter 11- 55.*

State of Hawaii, Department of Transportation, Airports Division. April 14, 2014. *National Pollutant Discharge Elimination System, Permit Number HI S000005*, expires March 13, 2019.

State of Hawaii, Department of Transportation, Airports Division. March 1, 2019. *Administrative Extension of National Pollutant Discharge Elimination System, Permit Number HI S000005*, expires on effective date of subsequent permit.

Appendix A

Stormwater Monitoring Location Figures

Appendix B
Blank DMR Form

Appendix C

Stormwater Sampling Log